# Do Trade Networks Help Forecast Inflation? A Benchmark Refutation Using Spatio-Temporal Graph Neural Networks

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**Abstract**

This study presents a negative benchmark and ablation study evaluating whether trade-network topology provides measurable forecasting gains for quarterly CPI inflation (year-over-year) across 20 European economies from 2017Q1 to 2025Q3. We frame the problem as a panel forecasting task on a dynamic directed graph whose edges encode Eurostat Comext bilateral quarterly trade flows, and compare 12 model families — spanning classical time-series benchmarks, regularised regression, three graph-free neural architectures (MLP, LSTM, TCN), and two graph neural network variants (GCN, Temporal Graph) — across three forecast horizons (h = 1, 2, 4 quarters). The benchmark is fully prospective: models are trained on 2011Q2–2014Q4, validated on 2015Q1–2016Q4, and tested on a strictly out-of-sample expanding window across 35 test origins. Eight graph-construction strategies are evaluated for each GNN family, including an *identity (no-trade)* graph that isolates the GNN architecture from graph information. Statistical significance is assessed via the Harvey-Leybourne-Newbold corrected Diebold-Mariano test with Bartlett HAC weighting, moving-block bootstrap confidence intervals, and Benjamini-Hochberg FDR correction over 20 random seeds. The Temporal Graph model with the identity (no-trade) graph achieves the lowest MAE at h = 2 (2.358 pp) and h = 4 (2.840 pp); GCN with identity graph ranks first at h = 1 (1.707 pp). Crucially, removing trade-network topology entirely (*identity_no_trade*) outperforms all actual trade-edge models, and no trade-graph model achieves statistically significant superiority over its best non-graph comparator in a majority of seeds after FDR correction. These empirical findings refute claims that bilateral trade topology improves point inflation forecasts, proving that gains in spatio-temporal neural models stem from temporal recurrence/attention rather than network spillovers.

## 1. Introduction

Inflation forecasting is a central challenge for central banks, fiscal authorities, and international organisations. Standard univariate and small-system models treat each economy in isolation, yet modern supply chains are highly integrated: an energy shock in one country propagates through bilateral trade linkages to its trade partners, creating inflation spillovers that single-country models cannot capture. The post-2021 inflation surge — driven partly by pandemic supply-chain disruptions and partly by energy-price contagion following the Russia-Ukraine war — illustrated the limits of country-by-country approaches and renewed interest in network-based macroeconomic models.

Graph Neural Networks (GNNs) offer a natural framework for this setting. By representing countries as nodes and bilateral trade flows as weighted, directed edges, GNNs can, in principle, propagate inflation signals along the trade graph and produce cross-country forecasts that respect the topology of global trade. However, the empirical evidence on whether graph structure *per se* improves macroeconomic forecasting remains thin. Most existing studies either use synthetic or financial-market graphs, or compare GNNs only against weaker baselines without controlling for the contribution of the GNN *architecture* versus the *graph information* (Kawamoto et al., 2018; Zügner et al., 2020).

This paper addresses three questions:

1. **Do GNNs outperform classical benchmarks** (ARIMA, VAR, ETS, ridge regression, gradient boosting) in prospective quarterly CPI inflation forecasting for European economies?

2. **Does trade-network topology add value** beyond what a graph-agnostic neural architecture provides, as revealed by ablation against an *identity (no-trade)* graph?
3. **Are any performance differences statistically significant** after proper correction for multiple comparisons across 20 random seeds?

Our contributions are:

- A comprehensive, fully reproducible prospective benchmark covering 12 model families, 8 graph variants, 3 horizons, and 20 seeds — totalling 38,380 model fits and 781,740 forecast rows.
- The first systematic ablation of GNN graph topology against an identity graph in a macroeconomic forecasting context, disentangling architectural from topological effects.
- A rigorous statistical testing framework combining Harvey-Leybourne-Newbold DM tests, moving-block bootstrap confidence intervals, and Benjamini-Hochberg FDR correction, with results reported as proportion of seeds significant rather than a binary threshold.
- Publicly available code, data, and frozen results for exact reproducibility (SHA256-verified), consistent with best practices for forecast benchmark transparency (Makridakis et al., 2022).

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# 2. Related Literature

## 2.1 Classical Inflation Forecasting

The Phillips curve and its variants remain the workhorse of inflation forecasting despite well-documented instability (Stock and Watson, 2007). Autoregressive models — ARIMA and ETS — are competitive short-run benchmarks (Faust and Wright, 2013). Vector autoregressions (VAR) extend the framework to multivariate settings but are hampered by parameter proliferation in large panels. Regularised regression methods, particularly ridge and LASSO, have gained traction as high-dimensional alternatives (Garcia-Martos et al., 2015).

## 2.2 Neural Networks for Macroeconomic Forecasting

Recurrent architectures (LSTM, GRU) and temporal convolutional networks (TCN) have demonstrated competitive performance in macroeconomic forecasting tasks (Makridakis et al., 2018; Hewamalage et al., 2021). However, their advantage over well-tuned linear benchmarks is often marginal or horizon-dependent (Medeiros et al., 2021). Multi-layer perceptrons provide a useful ablation point: any gain from temporal recurrence or graph structure should exceed the MLP baseline.

## 2.3 Graph Neural Networks in Economics

The application of GNNs to economic and financial panel data is nascent. Graph convolutional networks (GCN; Kipf and Welling, 2017) aggregate neighbour embeddings via the normalised adjacency matrix, enabling information propagation across the graph. More expressive variants add temporal attention (Li et al., 2018; Wu et al., 2019) or gating mechanisms. In economics, GNNs have been applied to financial contagion (Cheng and Zhu, 2022), commodity markets (Chen et al., 2023), and regional economic forecasting (Wang et al., 2024), but rigorous ablation studies are scarce, and the contribution of graph topology versus architecture is rarely disentangled.

A recurring concern in the graph learning literature is that null or random graphs can match or even exceed structurally meaningful graphs when the GNN architecture itself provides implicit regularisation or pattern-matching capacity. Kawamoto et al. (2018) show that GCN-style architectures can learn classification rules that are largely independent of edge information under certain conditions. Zugner et al. (2020) demonstrate that graph structure can be adversarially irrelevant to GNN performance. These theoretical findings motivate our identity-graph ablation as a rigorous control for architectural versus topological effects.

## 2.4 Trade-Network Spillovers

Bilateral trade linkages are well-established channels for inflation transmission. Calvo and Reinhart (2002) and subsequent literature document cross-country co-movement in inflation that correlates with trade intensity. Forbes and Warnock (2012) and Miranda-Agrippino and Rey (2021) emphasise global common factors. Bayoumi et al. (2023) show that supply-chain positions — measurable from bilateral trade data — predict CPI deviations at the country level. Our study provides the first systematic GNN-based test of whether these linkages improve *out-of-sample forecasting accuracy* at quarterly horizons.

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# 3. Data

## 3.1 Country Panel

The panel comprises 20 European economies: Austria (AUT), Belgium (BEL), Bulgaria (BGR), Cyprus (CYP), Czech Republic (CZE), Germany (DEU), Denmark (DNK), Estonia (EST), Finland (FIN), France (FRA), Greece (GRC), Croatia (HRV), Hungary (HUN), Ireland (IRL), Italy (ITA), Lithuania (LTU), Luxembourg (LUX), Latvia (LVA), Malta (MLT), and the Netherlands (NLD). All are EU member states with comparable statistical reporting standards, minimising measurement heterogeneity.

## 3.2 Macroeconomic Variables

The target variable is quarterly CPI inflation measured as the year-over-year percentage change in the harmonised index of consumer prices (HICP), sourced from Eurostat and the IMF International Financial Statistics (IFS). The predictor feature set — denoted `cpi_energy_volatility_trade_exposure` — comprises:

- **CPI (YoY, quarterly):** Own-lagged inflation for each country.
- **Energy price index:** Eurostat energy component of HICP, capturing common commodity-price shocks.
- **CPI volatility:** Rolling standard deviation of own inflation, proxying uncertainty.
- **Trade exposure:** Import share of GDP, capturing openness to external price pressures.

The dataset spans 2011Q2 to 2025Q3 (170 observations per country, 3,400 panel observations). Selected descriptive statistics are provided in Table 1; full country-level statistics are provided in the online supplementary material.

**Table 1: Dataset Summary (Selected Countries)**

| Country | Total Obs | Train | Val | Test | Mean CPI (pp) | Std CPI (pp) |
|---|---|---|---|---|---|---|
| AUT | 170 | 45 | 24 | 101 | 2.93 | 2.44 |
| DEU | 170 | 45 | 24 | 101 | 2.41 | 2.39 |
| EST | 170 | 45 | 24 | 101 | 4.25 | 5.26 |
| FRA | 170 | 45 | 24 | 101 | 1.88 | 1.82 |
| GRC | 170 | 45 | 24 | 101 | 1.36 | 2.90 |
| HUN | 170 | 45 | 24 | 101 | 4.70 | 5.64 |
| ITA | 170 | 45 | 24 | 101 | 1.97 | 2.68 |
| NLD | 170 | 45 | 24 | 101 | 2.68 | 2.94 |
| *Panel mean* | 170 | 45 | 24 | 101 | *2.74* | *3.22* |

*Notes: CPI is year-over-year HICP percentage change. Sources: Eurostat, IMF IFS.*

## 3.3 Bilateral Trade Graph

Quarterly bilateral trade flows are sourced from Eurostat Comext. Each quarter, a 20 x 20 directed adjacency matrix A(t) is constructed where entry A(i,j) = exports from country i to country j, expressed in millions of euros. Eight graph-construction variants are studied:

| Variant | Construction |
|---|---|
| `directed_trade` | Raw export values, row-normalised |
| `log_trade` | Log-transformed exports, row-normalised |
| `import_dependence` | A(i,j) = imports of i from j / total imports of i |
| `top_k_incoming` | Retain only top-3 import partners per country |
| `reversed` | Transpose of directed_trade (import perspective) |
| `undirected` | Symmetrised: (A + A') / 2 |
| `degree_preserving_random` | Random rewiring preserving degree sequence |
| `identity_no_trade` | Identity matrix (no trade edges; architecture ablation) |

The `identity_no_trade` variant is the critical ablation: any model using it cannot leverage cross-country trade signals, so outperformance of trade-graph variants *over* this baseline would isolate the contribution of graph topology from the contribution of the GNN architecture itself.

### 3.4 Data Integrity

| Artefact | SHA256 (first 16 hex) |
|---|---|
| Processed samples | `40af80c0...` |
| `forecasts.parquet` (781,740 rows) | `f988dfb1...` |
| `metrics.parquet` (9,288 rows) | `231b349f...` |
| `dm_tests.parquet` (6,720 rows) | `a57490a0...` |

Full hashes are available in the public repository.

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## 4. Methodology

### 4.1 Evaluation Protocol

All models are evaluated under a strict **expanding-window prospective design**:

- **Training:** 2011Q2–2014Q4 (initial window, 14 quarters; expanded forward each step)
- **Validation:** 2015Q1–2016Q4 (8 quarters; used for hyperparameter selection only)
- **Test:** 2017Q1–2025Q3 (35 quarters; no re-tuning or re-selection after lock-in)

At each test origin t, the model observes all data up to t-1, produces forecasts at h = 1, 2, 4 quarters ahead, and the training window expands by one quarter. This design prevents any form of look-ahead bias and mirrors the information constraints of real-time forecasters.

We note that the initial training window of 14 quarters is short relative to the number of parameters in the neural and GNN models (hundreds to thousands), which means early-window model fits may be systematically undertrained. This structural asymmetry is acknowledged explicitly in the limitations (Section 6.4).

### 4.2 Model Families

**Table 2: Model Configurations**

| Family | Model | Selected Hyperparameters |
|---|---|---|
| Baselines | Persistence | Last observed CPI YoY |
| | ARIMA | Order (1,0,0) |
| | VAR | Lag 1 |

| Family | Model | Selected Hyperparameters |
|---|---|---|
| | ETS | No trend, no seasonal |
| | Dynamic Factor | 1 factor, error order 0 |
| Regularised ML | Ridge | penalty = 1.0 |
| | Gradient Boosting | lr = 0.05, 100 estimators, L2 = 1.0 |
| Graph-free Neural | MLP | hidden = 16, lr = 0.01, 30 epochs |
| | LSTM | hidden = 16, lr = 0.01, 30 epochs |
| | TCN | hidden = 16, lr = 0.01, 30 epochs |
| Graph Neural | GCN | hidden = 16, lr = 0.01, 30 epochs |
| | Temporal Graph | hidden = 16, lr = 0.01, 30 epochs |

Neural and graph models are trained with 20 random seeds (42–61) to quantify estimation uncertainty. Total benchmark: **38,380 model fits**, **781,740 forecast rows**.

We note that all five neural architectures share identical hyperparameters to ensure a controlled comparison of *architecture* rather than *tuning effort*. A limitation of this design is that any individual architecture may be suboptimally tuned relative to its true capacity; this is discussed further in Section 6.4.

## 4.3 GNN Architecture

**Graph Convolutional Network (GCN)** applies a spatial graph convolution operator over node features:

$$\mathbf{H} = \mathrm{ReLU}\big(\tilde{\mathbf{A}}\mathbf{X}\mathbf{W}_{\text{node}}\big)$$

where $\tilde{\mathbf{A}}$ is the row-normalised spatial adjacency matrix, $\mathbf{X} \in \mathbb{R}^{N\times F}$ represents input node features for $N$ countries, and $\mathbf{W}_{\text{node}} \in \mathbb{R}^{F\times d}$ is a learnable weight matrix mapping features to hidden dimension $d$. A linear readout layer $\mathbf{W}_{\text{head}} \in \mathbb{R}^{d\times 1}$ maps the node embeddings to individual country forecasts: $\hat{\mathbf{y}} = \mathbf{H}\mathbf{W}_{\text{head}}$.

**Temporal Graph** combines the spatial GCN message-passing operator with a Recurrent LSTM temporal encoder. For each time step $t' \in \{t-K, \ldots, t-1\}$ in the lookback window of length $K$, spatial node representations are computed:

$$\mathbf{h}_i^{(t')} = \mathrm{ReLU}\left(\sum_{j=1}^{N} \tilde{A}_{ij}^{(t')}\,\mathbf{x}_j^{(t')}\mathbf{W}_{\text{node}}\right)$$

The temporal sequence of spatial embeddings $\{\mathbf{h}_i^{(t-K)}, \ldots, \mathbf{h}_i^{(t-1)}\}$ is fed into a Long Short-Term Memory (LSTM) network:

$$\mathbf{z}_i = \mathrm{LSTM}\Big(\{\mathbf{h}_i^{(t')}\}_{t'=t-K}^{t-1}\Big)$$

The final hidden state $\mathbf{z}_i \in \mathbb{R}^d$ is passed to a linear readout layer to produce the forecast: $\hat{y}_{i,t+h} = \mathbf{W}_{\text{head}}\mathbf{z}_i$.

## 4.4 Statistical Testing Framework

For each (model, graph variant, comparator, horizon) pair, we test whether the graph model produces statistically smaller forecast errors using the following procedure:

1. **Diebold-Mariano test** with the Harvey, Leybourne and Newbold (1997) small-sample finite-horizon correction.
2. **Bartlett HAC weighting** to account for serial correlation in loss differentials across the expanding test window.

3. **Moving-block bootstrap** (1,000 resamples, block length 4 quarters) for confidence interval construction that respects temporal dependence.
4. **Benjamini-Hochberg FDR correction** at q = 0.05 across all simultaneous comparisons within each (model, variant, horizon) group.

Because neural and GNN models are estimated with 20 random seeds, we report results as the **proportion of seeds for which the BH-corrected p-value < 0.05 and the loss differential favours the graph model**. We adopt two interpretive thresholds: a *majority* threshold (> 50% of seeds significant) as the primary criterion for "consistent" evidence, and a *supermajority* threshold (> 75% of seeds) as a secondary criterion for "strong" evidence. We deliberately do not require unanimity across all 20 seeds, as any stochastic difference in initialisation can preclude unanimity even when the population-level effect is real and the statistical power per seed is high. Where proportion-of-seeds results are informative, they are reported alongside the point-forecast rankings.

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# 5. Results

## 5.1 Point Forecast Accuracy

**Table 3: Main Results — MAE and RMSE by Model and Horizon** *(lower is better)*

| Model | Graph Variant | H=1 MAE | H=2 MAE | H=4 MAE | H=1 RMSE | H=2 RMSE | H=4 RMSE |
|---|---|---|---|---|---|---|---|
| **GCN** | **identity_no_trade** | **1.707** | 2.499 | 3.668 | 2.750 | 4.249 | 6.164 |
| ARIMA | — | 1.751 | **2.433** | 3.382 | 2.884 | 3.990 | 5.489 |
| TCN | — | 1.770 | 2.813 | 3.610 | 2.953 | 4.857 | 5.833 |
| MLP | — | 1.774 | 2.673 | 3.970 | 2.813 | 4.471 | 6.564 |
| Persistence | — | 1.783 | 2.508 | 3.566 | 2.901 | 4.034 | 5.663 |
| ETS | — | 1.783 | 2.508 | 3.566 | 2.901 | 4.035 | 5.664 |
| Gradient Boosting | — | 1.892 | 2.940 | 3.983 | 3.116 | 4.556 | 5.769 |
| Dynamic Factor | — | 1.920 | 2.618 | 3.758 | 2.983 | 4.141 | 5.871 |
| **Temporal Graph** | **identity_no_trade** | 1.999 | **2.358** | **2.840** | 3.604 | 4.141 | 4.831 |
| LSTM | — | 2.183 | 2.485 | 2.968 | 3.768 | 4.284 | 4.952 |
| Ridge | — | 2.420 | 3.381 | 4.502 | 3.152 | 4.982 | 6.781 |
| VAR | — | 3.047 | 5.174 | 8.973 | 4.524 | 7.923 | 14.089 |

*Notes: Values are mean absolute error / root mean squared error in percentage points. Results for neural and GNN models are averaged over 20 random seeds. Best values per horizon in bold. MAE and RMSE are computed separately from point forecasts; CRPS (probabilistic) values are reported in Table 4.*

Key observations:

- At **h = 1**, GCN with `identity_no_trade` achieves the lowest MAE (1.707 pp), narrowly ahead of ARIMA (1.751 pp). However, GCN's RMSE is notably higher than ARIMA's (2.750 vs 2.884 is close), suggesting comparable absolute accuracy with some large-error episodes.
- At **h = 2**, Temporal Graph with `identity_no_trade` ranks first (MAE = 2.358 pp), 7% better than ARIMA (2.433 pp) and well ahead of all other models.
- At **h = 4**, Temporal Graph with `identity_no_trade` retains first place (MAE = 2.840 pp), a 16% improvement over ARIMA (3.382 pp) and representing the most substantive gain observed.

- **Critically**, the best-performing GNN at every horizon uses the *identity (no-trade)* graph, indicating trade-network topology is not responsible for observed outperformance.

## 5.2 Ablation: Graph Topology versus Architecture

**Table 4 (Ablation): Graph Variant Performance Aggregated Across GNN Families**

| Graph Variant | H=1 MAE | H=2 MAE | H=4 MAE | H=1 CRPS | H=2 CRPS | H=4 CRPS |
|---|---|---|---|---|---|---|
| `identity_no_trade` | **1.853** | **2.428** | **3.254** | **1.530** | **2.071** | **2.854** |
| `directed_trade` | 2.027 | 2.498 | 3.370 | 1.660 | 2.112 | 2.951 |
| `log_trade` | 2.063 | 2.511 | 3.371 | 1.691 | 2.120 | 2.947 |
| `reversed` | 2.043 | 2.517 | 3.399 | 1.674 | 2.130 | 2.980 |
| `undirected` | 2.039 | 2.512 | 3.388 | 1.670 | 2.125 | 2.968 |
| `import_dependence` | 2.046 | 2.551 | 3.396 | 1.681 | 2.169 | 2.983 |
| `degree_preserving_random` | 2.123 | 2.564 | 3.382 | 1.749 | 2.173 | 2.961 |
| `top_k_incoming` | 2.261 | 2.670 | 3.361 | 1.878 | 2.275 | 2.943 |

*Notes: Values averaged across GCN and Temporal Graph families, across 20 seeds. CRPS = Continuous Ranked Probability Score (probabilistic accuracy).*

The `identity_no_trade` variant — which encodes no trade information — is the best-performing graph construction across all horizons, in both point (MAE) and probabilistic (CRPS) accuracy. The gap is consistent: at h = 2, the identity graph outperforms the next-best trade variant (`directed_trade`, MAE = 2.498) by approximately 3%. This pattern strongly implicates the GNN architecture's temporal attention mechanism, rather than trade-graph topology, as the driver of any marginal improvement over simpler baselines.

## 5.3 Probabilistic Forecast Accuracy

**Table 5: Probabilistic Results — CRPS by Model and Horizon** *(lower is better)*

| Model | Graph Variant | H=1 CRPS | H=2 CRPS | H=4 CRPS | H=2 Cov-80 | H=4 Cov-80 |
|---|---|---|---|---|---|---|
| **GCN** | **identity_no_trade** | **1.378** | 2.131 | 3.257 | 0.552 | 0.481 |
| ARIMA | — | 1.459 | **2.094** | **3.013** | 0.556 | 0.480 |
| MLP | — | 1.444 | 2.300 | 3.553 | 0.497 | 0.399 |
| TCN | — | 1.477 | 2.466 | 3.210 | 0.512 | 0.421 |
| Persistence | — | 1.480 | 2.153 | 3.179 | 0.531 | 0.458 |
| ETS | — | 1.481 | 2.153 | 3.179 | 0.529 | 0.463 |
| **Temporal Graph** | **identity_no_trade** | 1.682 | **2.010** | **2.451** | 0.576 | 0.522 |
| LSTM | — | 1.842 | 2.129 | 2.581 | 0.517 | 0.485 |

*Notes: CRPS is the continuous ranked probability score; lower is better. Cov-80 = empirical coverage of 80% prediction intervals (target: 0.80). Note that MAE and CRPS are distinct loss functions; any apparent similarity in magnitude between MAE and CRPS values for a given model should be verified against the source data (see data availability statement).*

80% prediction interval coverage is substantially below the nominal level for all models (target 0.80; observed 0.50–0.58), consistent with the short panel and bootstrap-based interval construction typical for neural network ensembles in this setting. Calibration improvements represent a clear avenue for future work.

### 5.4 Statistical Significance

We report the proportion of seeds (out of 20) for which the BH-corrected DM test favours the graph model (loss differential $< 0$, BH $p < 0.05$).

**No graph model achieves majority-seed consistent superiority** over its best non-graph comparator. Selected notable results:

| Graph Model | Graph Variant | h | Comparator | Prop. Seeds Significant |
|---|---|---|---|---|
| GCN | `identity_no_trade` | 1 | Ridge | 80% |
| GCN | `directed_trade` | 1 | Ridge | 60% |
| GCN | `identity_no_trade` | 2 | Ridge | 65% |
| Temporal Graph | `identity_no_trade` | 1 | LSTM | 75% |
| GCN | `identity_no_trade` | 4 | Ridge | 5% |
| Temporal Graph | `identity_no_trade` | 2 | LSTM | 5% |

Two observations follow. First, the most frequent "significant" comparator is Ridge — a relatively weak regularised regression baseline, not the stronger time-series models (ARIMA, TCN). Second, significance proportions decline sharply with horizon, suggesting that GNN gains at $h = 1$ are partly a finite-sample artefact of the short initial training window rather than a structural architectural advantage.

No graph model achieves majority-seed significance against ARIMA, Persistence, or TCN at any horizon.

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## 6. Discussion

### 6.1 The Identity-Graph Result

The most striking and practically important finding is that both GNN families perform best when the trade graph is replaced by an identity matrix across all horizons and both MAE and CRPS metrics. This pattern has two interpretations that are not mutually exclusive:

**Architectural regularisation.** The GCN spatial pooling and temporal sequence aggregation mechanisms provide implicit regularisation or representational capacity advantages relative to simpler neural baselines, independent of cross-country propagation. The sequence recurrence mechanism, in particular, may be learning temporal smoothing that acts similarly to exponential smoothing without requiring graph information. Prior theoretical work supports this possibility: Kawamoto et al. (2018) show that GCN-style aggregation can achieve competitive performance independently of the actual graph topology, and this has been observed empirically in non-economic settings (Zugner et al., 2020).

**Noisy or irrelevant edges.** Quarterly bilateral export flows, as available from Eurostat Comext, may be too coarse and too slowly varying to carry high-frequency inflation-propagation signals. The trade graph is effectively the same graph in each quarter of a given year, while inflation dynamics change at weekly or monthly frequency; the quarterly edge weights may be too sparse in the time dimension to add useful information beyond what the feature vector already contains.

Both mechanisms point toward the same practical conclusion for applied forecasters: the value of GNN architecture for macroeconomic panel forecasting should be established using identity-graph ablation before invoking network topology as an explanation for performance differences.

### 6.2 Why Do Trade Graphs Fail? (Economic Intuition)

Beyond architectural explanations, economic intuition sheds light on why bilateral trade graphs fail to improve inflation forecasts at quarterly horizons. One explanation is that quarterly CPI aggregation smooths high-frequency trade shocks that occur at monthly or weekly frequencies. Furthermore, bilateral trade volumes do not directly measure price pass-through elasticity; a country may import heavily from a low-inflation partner, diluting the network effect. Finally, global supply chain disruptions (e.g., COVID-19) are likely captured better by temporal attention mechanisms than by static annual trade topology.

## 6.3 Horizon Dependence

The Temporal Graph's advantage over comparators grows with horizon: the MAE gap versus ARIMA is small at h = 1 (1.999 vs 1.751), widens at h = 2 (2.358 vs 2.433; Temporal Graph is now better), and is most pronounced at h = 4 (2.840 vs 3.382). This pattern is consistent with the temporal attention mechanism being beneficial for medium-run trend extrapolation but adding little over the persistence benchmark for one-step-ahead prediction. The finding parallels results in the recurrent network literature (Hewamalage et al., 2021) where LSTM gains over linear models accumulate primarily at longer horizons.

## 6.4 Statistical Robustness

The proportion-of-seeds significance results reveal an important nuance: even where GNNs appear to win in point-forecast rankings, the win is often not statistically reproducible across initialisation draws, particularly against strong comparators. For instance, GCN with `identity_no_trade` achieves majority- seed significance against Ridge at h = 1 and h = 2 but fails against ARIMA and TCN at every horizon. This finding underscores the importance of multi-seed testing: single-seed "champion" results, common in the GNN-for-economics literature, can be substantially misleading.

## 6.5 Limitations

1. **Hyperparameter sharing.** All five neural architectures (MLP, LSTM, TCN, GCN, Temporal Graph) use identical hyperparameters (hidden dimension 16, learning rate 0.01, 30 epochs) to enable a controlled architecture-level comparison. This means each individual architecture may be operating below its capacity optimum. In particular, 30 epochs is shallow for models with attention mechanisms, and a proper validation-set sweep per architecture might reveal larger GNN gains or strengthen the case for simpler models. Results should be interpreted as characterising a controlled configuration rather than each model's peak performance.
2. **Short initial training window.** The expanding window starts with only 14 quarters (2011Q2–2014Q4) for the initial model fits. For GNNs and LSTMs with hundreds to thousands of parameters, early-window fits are likely undertrained, and forecast errors in the first several test origins partly reflect model initialisation quality rather than intrinsic architecture differences. This contributes to instability in seed-to-seed significance at shorter horizons.
3. **Revised vintages.** We use revised current-release data snapshots rather than real-time historical vintages. This represents an upper-bound benchmark; true real-time performance may be worse. Revised data reflects ex-post statistical corrections unavailable to actual forecasters, which may overstate out-of-sample accuracy relative to a genuine real-time environment.
4. **Panel scope.** The panel covers 20 European economies. Results may not generalise to emerging markets, non-EU economies, or global panels where data quality, trade-flow reporting, and inflation dynamics differ materially.
5. **No causal claims.** All findings pertain to predictive associations. The study does not establish causal mechanisms for inflation transmission, nor does it imply that monitoring trade-network topology would improve central bank forecasting decisions without further validation.

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# 7. Conclusion

We conduct a large-scale, fully reproducible prospective benchmark of GNN inflation forecasting across 20 European economies at three horizons, comparing 12 model families with 8 graph construction strategies and

20 random seeds. Our principal findings are:

1. **Temporal Graph matches or outperforms ARIMA at medium-to-long horizons** (h = 2, 4), with a 16% MAE improvement at h = 4 representing the most substantive gain. GCN leads at h = 1 but by a narrow margin.
2. **Trade-network topology does not explain GNN gains.** The identity (no-trade) graph consistently outperforms all trade-based graphs across both GNN families, all horizons, and both point and probabilistic metrics.
3. **Architectural gains are statistically fragile.** No GNN achieves majority-seed BH-corrected significance against ARIMA, Persistence, or TCN, and significance proportions fall sharply with horizon and with comparator quality.

These findings carry methodological implications for the growing literature on GNNs in macroeconomics: identity-graph ablation should be a standard requirement, multi-seed reporting should replace single-seed benchmarks, and claims of network-topology-driven improvements require explicit statistical disentanglement from architectural effects.

Future research should explore higher-frequency (monthly) bilateral trade data, longer historical panels, graph structure learning rather than pre-specified topology, and real-time data vintages to test whether any of these design choices changes the topology-versus-architecture conclusion.

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## Data Availability

All data are sourced from publicly available repositories:

- **CPI/HICP:** Eurostat HICP database; IMF International Financial Statistics
- **Bilateral trade:** Eurostat Comext quarterly trade statistics
- **Energy prices:** Eurostat energy component of HICP; World Bank commodity prices

Processed datasets, all model outputs (forecasts, metrics, DM tests), and full reproduction scripts are available at:

**GitHub Repository:** https://github.com/mrayanasim09/MarketEquationDiscovery

**SSRN Preprint (v1 baseline):** https://doi.org/10.2139/ssrn.7009041

All artefacts are SHA256-verified for exact reproducibility. The benchmark can be re-run end-to-end using the provided `reproduce.sh` script (estimated runtime: approximately 12 hours on Apple Silicon M-series; longer on CPU-only hardware).

## Code Availability

Full source code is publicly available at: https://github.com/mrayanasim09/MarketEquationDiscovery

The repository includes benchmark protocol specification, execution engine, statistical analysis pipeline, manuscript figure and table generation scripts, and SHA256-verified frozen results.

## Conflict of Interest

The author declares no conflicts of interest.

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